Identify or Certify? What Public Market Data Reveal About Algorithmic Feedback

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Abstract

Regulators ask to know which algorithms are trading. We ask which questions about those algorithms market data can answer, and at what cost. In a closed-loop model in which cohorts of algorithms respond to prices with exponential memory, every question an outside observer can ask is a functional of the merged measure of the cohorts' memory rates and weights. Identifying the cohorts is expensive: separating K cohorts whose memory rates are $\Delta\eta$ apart needs a number of observations of order $(\Delta\eta)^{-(4K-2)}$, from controlled experiments and from passive data alike. Certifying the market's safety is cheap: it depends on the cohorts only through one linear functional, the stability margin $M = 1-a-\kappa\Phi$, and from passive data deciding its sign takes a record of length of order $1/M$: we prove lower and upper bounds of that order for one closed-loop root and for any number of cohorts with finite memory. The two costs are unrelated: one depends on how similar the agents are, the other on how close the market is to instability. Three tests registered before their data were downloaded, each on 25 to 30 crypto perpetual futures never used before, support the dichotomy: the margin was decided, always on the safe side, in 22 of 30, 23 of 25 and 24 of 26 markets, while the cohorts were identified in at most one per test. The certificate is informative about turbulence but not about crashes: on 34 further new markets it forecast next-day variance beyond the Hawkes branching ratio, as registered ($t = 4.84$), while registered forecasts of adverse and extreme moves failed and it did not rank which markets fell

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furthest on the worst days; on 36 more, losses scaled with the estimated closed-loop response at about half the predicted rate. With contrarian agents, which dominate these data, the theory predicts exactly this: the margin certifies stability, but it stops measuring worst-case loss. We prove that the quantity that does, the horizon-limited absolute response, is certified from passive data at the parametric rate, that the cost of order $1/M$ holds uniformly as the margin vanishes, and that without a bound on the agents' memory no passive certificate exists at all.

Keywords: algorithmic trading, identification, certification, closed-loop markets, unit roots, pre-registration. **JEL:** G14, G18, C12, C22.

1 Introduction

When trading algorithms condition on recent prices and their orders move prices, price formation is a feedback system. Regulators increasingly respond by asking to know which algorithms are trading: MiFID II requires firms to flag algorithmic orders and identify the algorithm responsible, and the US Consolidated Audit Trail links orders to customers. The premise is that knowing the agents is the route to knowing whether the market is safe. This paper shows that, for a supervisor who only sees market data, the two questions have very different costs, and that the cheap one is the one that matters for safety.

We work with a linear closed-loop model in which K cohorts of algorithms track the price displacement with exponential memory, the reduced form of steady-state Kalman filtering and of regularized online learning, and trade on it with weights that may be positive (momentum) or negative (contrarian). The agents enter the observable dynamics only through a merged measure of memory rates and weights (Section 2). Every question about them is a functional of that measure, and we sort questions by their cost.

Identification is expensive. Separating two cohorts whose memory rates differ by δ needs $\Omega(\delta^{-4})$ observations, and separating K cohorts spaced $\Delta\eta$ apart needs $\Omega((\Delta\eta)^{-(4K-2)})$, whether the observer runs controlled experiments or watches passively (Section 3). These bounds are attained.

Certification is cheap. Safety depends on the agents only through their total feedback weight Φ, and the market is stable exactly when the margin $M = 1 - a - \kappa\Phi$ is positive. From reset experiments, certifying Φ costs $\sigma^2\epsilon^{-2}\log(1/\epsilon)$ and no less. From passive data, deciding the sign of M is a unit-root problem: we prove that it needs a record of length of order $1/M$, with lower and upper bounds of that order, first for one closed-loop root and then for any number of cohorts whose memory is finite (Theorems 5 and 6). The number and similarity of the cohorts do not enter.

The dichotomy. Together, the cost of identification is set by how similar the agents are and the cost of certification by how close the market is to the boundary, and for a market at a fixed margin the ratio of the two grows without bound as the agents become similar (Theorem 7). A supervisor can know that a market is stable long before it can know who trades in it.

Evidence. We test the dichotomy on public Binance data with three tests whose specifications, code hashes and universe rules were pushed to a public repository in commits of

their own before any of their data were downloaded, each run once (Section 7). On three disjoint sets of perpetual futures never used before, the margin was decided in 22 of 30 (one short of the registered threshold), 23 of 25 and 24 of 26 markets, always on the safe side, while the memories were identified in 1, 1 and 1. Markets in which the decision was reached but the agents were not identified outnumbered the reverse by 22 to 1, 22 to 0 and 23 to 0. Class-level positioning data, the closest public analogue of tagged disclosure, restricted a persistence parameter that prices could not bound in 29 of 29 markets.

What the certificate is not. The margin certifies distance from instability. It is not a forecast of the next crash. A registered test of whether the 20-day margin forecasts the next day's largest adverse move failed (Clark–West $t = 1.51$, $p = 0.065$). A later registered test on 34 new markets found that it does forecast next-day realized variance, beyond the Hawkes branching ratio ($t = 4.84$), but not extreme moves, and that it does not rank damage on market-wide stress days. A third registered test found that the size of the estimated closed-loop response does scale with next-day losses within markets, at about half the rate the linear theory predicts, but that beyond recent volatility it forecasts smaller losses, not larger. With contrarian agents, which the estimates suggest are common, the summed response no longer bounds worst-case loss (Proposition 8); we prove that the quantity that does is certified from passive data at the parametric rate, although at a one-hour horizon the data do not yet distinguish the two. Safety in the sense of stability is certifiable from prices without identifying anyone; turbulence is partly forecastable from the same certificate; crashes are not.

Related literature. The identification bounds belong to the theory of super-resolution and mixture estimation (Heinrich and Kahn, 2018; Wu and Yang, 2020); their market form, with the costs in executed quantity and capital, is in the companion paper (Niroula and Niroula, 2026a). The cost of a linear question is governed by a modulus of continuity, as in the classical theory of estimating linear functionals (Ibragimov and Has'minskii, 1984; Donoho, 1994); what is specific to markets is that the safety functional sits exactly on the boundary of the parametric class. Passive certification is a unit-root problem, and we use the limit theory of Mann and Wald (1943) and Phillips and Magdalinos (2007); the impossibility of certifying markets close to the boundary from short records is the reason unit-root tests have little power near unity (Elliott et al., 1996). The closest empirical measure of self-excitation in markets is the branching ratio of Hawkes models of order arrivals (Filimonov and Sornette, 2012; Hardiman et al., 2013), which estimates how much trading is triggered by trading; our margin measures the price feedback loop instead, and in a registered head-

to-head comparison the margin forecast next-day variance while the branching ratio did not (Section 7.5). On algorithmic trading and fragility, Brunnermeier and Pedersen (2009) and Kirilenko et al. (2017) study how feedback and liquidity interact in stress; Budish et al. (2015) study market design responses.

What we do not claim. The bounds hold for the stated linear model and observation contracts. They are not limits on what participant-level records can reveal; with such records identification becomes ordinary estimation (Niroula and Niroula, 2026a). The empirical margin is estimated from a reduced form whose relation to the structural model rests on the model's linearity and on the stationarity of the margin over the estimation window; we report where these assumptions strain.

2 Model

The displacement of the price from its fundamental value, x_t, evolves as

$$x_{t+1} = a\,x_t + \kappa\,(u_t + w_{t+1}), \qquad u_t = \sum_{i=1}^{K} q_i\,y_{i,t}, \qquad y_{i,t} = r_i\,y_{i,t-1} + (1-r_i)\,x_t, \tag{1}$$

with persistence $a \in [0,1)$, impact $\kappa > 0$, shocks w_t, and K cohorts with weights $q_i \in \mathbb{R}$ and memory rates $r_i \in [0,1)$; a cohort with $q_i < 0$ is contrarian. The aggregate response is a linear filter of past displacements,

$$u_t = \sum_{m \geq 0} k_m\,x_{t-m}, \qquad k_m = \int r^m\,dQ(r), \qquad Q = \sum_{i=1}^{K} q_i(1-r_i)\,\delta_{r_i},$$

so the agents enter only through the *merged measure* Q: cohorts that share a memory rate are indistinguishable whatever their identities. The total feedback weight is $\Phi = \sum_m k_m = \int (1-r)^{-1} dQ(r) = \sum_i q_i$, and the closed-loop polynomial $f(z) = 1 - az - \kappa z \sum_m k_m z^m$ has $f(1) = 1 - a - \kappa\Phi$.

Definition 1 (Stability margin)**.** The *stability margin* is $M = 1 - a - \kappa\Phi$. When every $q_i \geq 0$ (the positive class), the closed loop is stable if and only if $M > 0$, every tap of the impulse response h is non-negative, and the worst-case transient loss of a position I under shocks bounded by $\bar{w}$ is $|I|\bar{w}\|h\|_1 = |I|\bar{w}\,S$ with

$$S = \sum_t h_t = \frac{\kappa}{1 - a - \kappa\Phi} = \frac{\kappa}{M}.$$

Safety in the positive class is therefore a question about one linear functional of Q, while identification asks for Q itself. We compare the two under two observation contracts: *reset experiments*, in which an observer can restart the market from rest, apply a probe and read noisy responses; and *passive data*, in which the observer only watches the market run. The first is the contract of a supervisor with market access; the second is the contract of anyone with public data.

3 Identification is expensive

Theorem 2 (Merged equivalence and the cost of separation; Niroula and Niroula, 2026a)**.** *Let a and κ be known.*

(i) Two economies produce identical price responses to every probe if and only if their merged measures coincide.

(ii) With reset experiments whose noisy responses have variance σ^2, separating a single cohort at memory rate $\bar{r}$ from two cohorts at $\bar{r} \pm \delta$ with the same total weight needs $\Omega(\sigma^2\delta^{-4})$ experiments under an average-power constraint on probes; separating two K-cohort economies whose rates lie in an interval of width $2\Delta\eta$ and whose merged measures share their first $2K-1$ moments needs $N \geq c\,\sigma^2(\Delta\eta)^{-(4K-2)}$.

(iii) With passive data, the Kullback–Leibler divergence rate between the same pairs is $\Theta(\delta^4)$ and $\Theta((\Delta\eta)^{4K-2})$, so a passive record needs length $\Omega(\delta^{-4})$ and $\Omega((\Delta\eta)^{-(4K-2)})$, and the likelihood-ratio test attains these orders.

The bounds are proved in Niroula and Niroula (2026a), which also translates them into executed quantity of order δ^{-2} and trading capital of order δ^{-4} for a known pair. The point for this paper is part (iii): watching instead of probing does not make identification cheaper. The bounds do not depend on how close the market is to instability.

4 Certification is cheap

4.1 Reset experiments

Under the reset contract, a linear question $F_g(Q) = \int g\,dQ$ over measures supported on $[0, \bar{r}]$ with mass at most B costs a number of experiments set by a modulus of continuity whose dual is a problem of polynomial approximation with penalized coefficients, and it is parametric exactly when g has square-summable power-series coefficients (Niroula, 2026b,

Chapter 4). The total feedback Φ has $g(r) = 1/(1-r)$, whose coefficients are all one: it sits on the boundary.

Theorem 3 (The safety question costs one logarithm; Niroula, 2026b)**.** *Under the reset contract, with memory bound $\bar{r} < 1$, certifying Φ, and S whenever $a + \kappa B/(1-\bar{r}) < 1$, to half-width ϵ needs $N \asymp \sigma^2 \epsilon^{-2} \log(1/\epsilon)$ experiments as $\epsilon \to 0$, and the logarithm cannot be removed. Without a memory bound, no finite number of experiments certifies Φ to any finite accuracy.*

Corollary 4 (What the closed loop adds)**.** *In the positive class with $M > 0$ and $\epsilon \in (0, 1/2]$, certifying S to relative accuracy ϵ is equivalent, up to a factor of two in accuracy, to certifying Φ to half-width $\epsilon M/\kappa$; it costs of order $\sigma^2 \kappa^2 M^{-2} \epsilon^{-2} \log(\kappa/(\epsilon M))$ experiments.*

Proof. S is increasing in Φ on $\Phi < (1-a)/\kappa$. If $|\Phi' - \Phi| \le \epsilon M/(2\kappa)$ then $M' = 1 - a - \kappa\Phi' \ge M(1 - \epsilon/2)$ and $|S' - S|/S = \kappa|\Phi' - \Phi|/M' \le \epsilon$. Conversely $|S' - S| \le \epsilon S$ gives $\kappa|\Phi' - \Phi| \le \epsilon M' \le \epsilon(M + \kappa|\Phi' - \Phi|)$, so $\kappa|\Phi' - \Phi| \le 2\epsilon M$. Substitute into Theorem 3. □

The economics of the problem, how close the market is to instability, enters the cost of certification and only of certification.

4.2 Passive data

Public data are passive. Deciding the sign of the margin from them is the problem of rejecting a unit root.

Theorem 5 (Passive certification, one root)**.** *Observe $x_{t+1} = a\, x_t + \kappa \sum_{m \ge 0} k_m x_{t-m} + \kappa w_{t+1}$ with independent $N(0, s^2)$ shocks for $t < T$, from $x_0 = 0$.*

(i) For $\mu \in (0, 1]$, two economies with one cohort of memory rate zero and margins μ and $-\mu$ have Kullback–Leibler divergence at most $2T\mu$; deciding the sign with error at most $\alpha < 1/2$ under both needs $T \ge (1-2\alpha)^2/\mu$.

(ii) In these economies $x_{t+1} = \rho x_t + \kappa w_{t+1}$ with $\rho = 1 - M$. For fixed $\rho \in (-1, 1)$, $\sqrt{T}(\hat{M} - M) \to N(0, M(2-M))$ for the least-squares $\hat{M} = 1 - \hat{\rho}$; along $M_T \to 0$ with $TM_T \to \infty$, $(T/2M_T)^{1/2}(\hat{M} - M_T) \to N(0, 1)$. The studentized interval decides the sign with asymptotic error at most α once $TM > 2z^2_{1-\alpha/2}$.

Theorem 6 (Passive certification, several cohorts)**.** *Let the kernel vanish beyond $L \ge 2$ lags, with any signs, and write the economy as $x_{t+1} = \sum_{i<L} \phi_i x_{t-i} + e_{t+1}$ with e_t independent $N(0, \sigma^2)$ and $x_t = 0$ for $t \le 0$, so that $f(z) = 1 - \sum_{i<L} \phi_i z^{i+1}$ and $M = f(1)$. Suppose $f(z) = (1 - \rho z) g(z)$ with g real, $g(0) = 1$, of degree $p = L - 1$ and without zeros in $|z| \le 1 + \gamma$, $\gamma > 0$. Then $g(1) > 0$, $M = (1-\rho) g(1)$, and:*

(i) for $0 < \mu \le g(1)$ the economies with $\rho_\pm = 1 \mp \mu/g(1)$ have margins $\pm\mu$ and divergence at most $2T\mu/g(1)$; deciding the sign with error at most $\alpha < 1/2$ needs $T \ge (1 - 2\alpha)^2 g(1)/\mu$;

(ii) for $\rho \in [0, 1)$, the least-squares estimate of $-M$ in the exact augmented regression of $x_{t+1} - x_t$ on x_t and p lagged differences satisfies $\sqrt{T}(\hat{M} - M) \to N(0, V)$ with

$$V \le C_{p,\gamma} M, \qquad C_{p,\gamma} = 4p^2(p+1)(1+\pi^2)\, 4^p \big((1+\gamma)/\gamma\big)^p,$$

so the studentized interval has half-width below M once $T > z^2_{1-\alpha/2} C_{p,\gamma}/M$.

The proofs are in Appendix A. The constant in part (ii) is crude: in numerical checks with $p = 2$ and ρ up to 0.99, V/M is within 15% of $2g(1)$, so that the upper bound needs T of order $2z^2 g(1)/M$ and the lower bound $(1 - 2\alpha)^2 g(1)/M$, matching in their dependence on both $g(1)$ and M. Section 6 shows that the passive bounds need a memory bound and hold uniformly as $M \to 0$; the pooled price-and-flow estimator used in our tests is not covered.

Theorem 7 (Identify or certify)**.** *Under the passive contract, separating K cohorts whose memory rates are $\Delta\eta$ apart needs a record of length $\Omega((\Delta\eta)^{-(4K-2)})$ whatever the margin, while deciding the safety of a market with finite memory at margin M needs a record of length of order $1/M$ whatever the cohorts' spacing, with constants depending only on the memory length and the root gap. For a market at a fixed margin, the ratio of the two costs grows without bound as the cohorts become similar.*

Proof. Theorem 2(iii) and Theorem 6; the latter's constants do not involve the memory rates of the cohorts beyond the root gap of g, and its bounds do not involve $\Delta\eta$. □

4.3 A drifting margin

Estimated margins drift over months in our data while their sign stays put. If each day gives an unbiased reading $m_t = M_t + e_t$ with noise $N(0, s^2)$ and the margin moves by at most d per day, averaging the last W readings has bias at most $d(W+1)/2$ and standard deviation $s/\sqrt{W}$; the best window is $W^* = (zs/d)^{2/3}$, and no procedure, using any history, can certify margins below a floor of order $(s^2 d)^{1/3}$ (Niroula, 2026b, Proposition 4.7.4). Calibrated to quarter-by-quarter estimates on 27 spent markets, the best window is about 53 days, close to one quarter.

5 Contrarian agents and what the margin measures

The positive class assumes every agent follows the price. Our data say otherwise: the estimated feedback term $\kappa\Phi$ was negative in 14 of 30, 16 of 25 and 14 of 26 markets in the three tests.

Proposition 8 (Contrarian agents)**.** *Allow signed weights, so that the merged measure is signed with total variation at most B.*

(i) Linear questions, including Φ, cost at most a constant factor more than in the positive class: the modulus of continuity at most doubles.

(ii) The summed response no longer measures risk. The worst-case loss is $|I|\bar{w}\|h\|_1 \geq |I|\bar{w}|S|$, and $\|h\|_1/|S|$ is unbounded on stable economies with a memory bound: one contrarian cohort with memory rate zero and $a + \kappa q = -q_0$, $q_0 \in [0, 1)$, has $h_t = \kappa(-q_0)^{t-1}$, $|S| = \kappa/(1 + q_0)$ and $\|h\|_1 = \kappa/(1 - q_0)$.

(iii) The horizon-limited worst case $\bar{w}\|h\|_{1,H}$ is certified from reset experiments with no memory bound and no sign condition, at a cost of order $\sigma^2 H^2 \epsilon^{-2}$ instead of $\sigma^2 H \epsilon^{-2}$ in the positive class.

Proof. (ii) With memory rate zero, $k_0 = q$ and $h_{t+1} = (a + \kappa q)h_t$ with $h_1 = \kappa$; sum the geometric series. Parts (i) and (iii) are proved in Niroula (2026b, Proposition 4.9.1). □

The margin is the right statistic for whether a market is stable, in either class. It is the right statistic for worst-case loss only in the positive class. In markets where contrarian feedback is common, a low margin can coexist with an oscillating response whose absolute size the margin does not bound, and a high margin with large losses. Section 7 shows that this is what the data look like.

6 The price of memory, uniformity, and what to certify instead

Theorem 9 (Hidden slow feedback)**.** *Let $\mathcal{E}_0$ be the passive economy with one cohort of memory rate zero and margin $c \in (0, 1]$, and let $\mathcal{E}_r$ add a momentum cohort with memory rate $r < 1$ and weight $(c + \mu)/\kappa$, so that its margin is $-\mu < 0$. Over T observations from rest, the divergence between them is at most $(c + \mu)^2(1 - r)^2 T^3/(2c)$. Hence, for every T and every procedure, $\sup_{r<1} \Pr_{\mathcal{E}_r}(\textit{certify safe}) \geq \Pr_{\mathcal{E}_0}(\textit{certify safe})$: without a memory bound,*

passive certification has power at most its size, however safe the market. With memory bound $\bar{r}$, deciding the sign with error at most α needs $T \geq \big(2(1-2\alpha)\sqrt{c}/((c+\mu)(1-\bar{r}))\big)^{2/3}$.

Proof. The conditional means differ by $D_t = (c+\mu)(1-r)\sum_{m\leq t} r^m x_{t-m}$; by Cauchy–Schwarz and $\mathbb{E}_0 x_s^2 \leq \sigma^2/c$, $\mathbb{E}_0 D_t^2 \leq (c+\mu)^2(1-r)^2(t+1)^2\sigma^2/c$, and summing over $t < T$ gives the bound. Pinsker's inequality gives the rest. □

The theorem is the market form of the observation that unit-root tests have power no greater than size against all stationary processes (Blough, 1992; Faust, 1996; Pötscher, 2002). Its economic content is specific: a slow momentum cohort, trading on long averages of the price, adds little to each period's response and a great deal to total feedback, so it can make a market unstable without leaving a trace in a finite record. In simulation with $c = \mu = 0.2$ and $T = 1000$, an augmented Dickey–Fuller certificate certified the safe economy in 100% of replications and the unsafe economy with memory rate 0.999 in 100%. What disclosure must supply for safety is therefore not the agents' identities but a bound on how slowly the largest of them trade.

Theorem 10 (Uniform certification with finite memory)**.** *On the finite-memory economies of Theorem 6 with $\rho \in [0,1]$ and g in a compact set with root gap γ, the augmented Dickey–Fuller certificate at level α has size tending to α uniformly on the boundary $M = 0$, tends to zero for every fixed explosive economy, and for every $\epsilon > 0$ certifies safety with probability at least $1-\epsilon$, uniformly, once $TM/g(1) \geq C_\epsilon$. With Theorem 6(i), the record length needed is of order $g(1)/M$ uniformly as $M \to 0$.*

Proof. With $\theta = T(1-\rho) = TM/g(1)$, the t-statistic converges to the local-to-unity functional τ_θ of an Ornstein–Uhlenbeck process (Phillips, 1987; Chan and Wei, 1987), uniformly in $\theta \geq 0$ and in the nuisance (Mikusheva, 2007, 2012); τ_0 is the Dickey–Fuller law and $\tau_\theta \to -\infty$ as $\theta \to \infty$. □

In simulation the certification rate depends on the margin only through $TM/g(1)$: across two choices of g and $T \in \{500, 2000\}$ it was 4–7% at $\theta = 0$, 68–78% at $\theta = 10$ and at least 100% at $\theta = 40$.

The margin decides stability; with contrarian agents it does not bound loss. The horizon-limited absolute response does: with ψ_j the coefficients of $1/f$, a shock sequence of k innovation standard deviations moves the displacement by at most $k\sigma\Psi_H$ within H periods, $\Psi_H = \sum_{j<H} |\psi_j|$, and some sequence attains it.

Theorem 11 (Passive certificate of worst-case loss)**.** *On a compact set of stationary finite-memory economies with signed kernels, the interval $\Psi_H(\hat{\phi}) \pm G_H(\chi^2_{L,1-\alpha}\lambda_{\max}(\hat{\Sigma})/T)^{1/2}$, with*

G_H a Lipschitz constant of Ψ_H and $\hat{\Sigma}$ the estimated covariance of the least-squares coefficients, covers Ψ_H with probability at least $1-\alpha+o(1)$ uniformly; its half-width is of order $T^{-1/2}$, and no interval does better than order $T^{-1/2}$. The summed response $1/M$ can understate Ψ_∞ by an unbounded factor.

Proof. Uniform asymptotic normality of $\hat{\phi}$ on compact sets of stationary autoregressions (Mann and Wald, 1943; Mikusheva, 2007), the polynomial dependence of each ψ_j on ϕ, and $\big||u|-|v|\big| \le |u-v|$ give the upper bound. For the lower bound, $\Psi_H = \sum_{j<H} \rho^j$ in the memory-zero economies is strictly increasing with derivative at least one, and roots ρ and $\rho + \delta T^{-1/2}$ have divergence bounded in T. □

In a simulated contrarian economy the interval covered Ψ_H in 100% of replications, and the summed response understated it by 38%.

7 Evidence

7.1 Design

All tests use Binance USD-M perpetual futures from the public archive (`data.binance.vision`), one-minute klines with taker-buy volume, aggregated to five-minute marks. On each valid day the margin is estimated from two regressions whose sufficient statistics are pooled over days: five-minute returns on signed taker flow and the lagged displacement from an exponential moving average, which give κ and the reversion $-\beta = 1-a$; and flow on twelve lags of the displacement, which gives the kernel. The margin is $\hat{M} = -\hat{\beta} - \hat{\kappa}\hat{\Phi}$, decided if its 95% day-bootstrap interval excludes zero. The memories are identified if a two-exponential fit to the return kernel separates two rates with a bootstrap interval excluding zero and neither rate's interval touching a bound of the fit.

Each test's specification, the hashes of its code, its universe rule and its seed were pushed to a public repository in a commit of their own before any of its data were downloaded, and a counter that the code increments on opening prevents a second run. Universes are drawn from symbols that appear in no file of the repository, so every test uses markets never used before. Results are reported as they came out, and explanations found afterwards are labelled post hoc.

7.2 Certification without identification

Table 1 gives the result. With 105 days per market the margin was decided in 22 of 30, one short of the threshold; with 430 days, in 23 of 25 and 24 of 26, and on the first 105 days of

Table 1: Registered tests of the dichotomy

	C2	C3	F1
New markets drawn / evaluable	30 / 30	30 / 25	30 / 26
Days per market	105	430	430
Margin decided (registered threshold)	22 (23)	23 (19)	24 (20)
of which unsafe	0	0	0
Memories identified	1	1	1
Decided, not identified : identified, not decided	22 : 1	22 : 0	23 : 0
Same decision without the feedback term	26	22	22
Primary 1 (margin decided in $\geq 75\%$)	failed	passed	passed
Primary 2 (McNemar)	passed	passed	passed

C2 used 105 dates; C3 and F1 used 430 dates from January 2025 to September 2026 and C3's valid-day rule (at least half of five-minute returns non-zero; at least 200 valid days). Markets with fewer valid days were excluded under that rule. Holm correction across each test's primaries.

C3's markets in 20: the decision comes with the length of the record, as Theorem 5 predicts. Every decided margin was on the safe side. The memories were identified in at most one market per test. Quarter by quarter, in an exploratory analysis of 169 symbol-quarters, the margin was decided in 135, never on the unsafe side.

Two qualifications matter. The decision rests mainly on reversion: without the feedback term it was the same in most markets, so these data certify that the markets are far from the boundary without measuring the feedback precisely. And the theory's prediction that the standardized margin grows like the square root of the record holds across markets but not market by market, because the margin drifts.

7.3 Disclosure

Binance publishes five-minute long/short positioning ratios for a top-trader class and for all accounts. In C2 a registered primary compared the width of the interval for the class persistence rate with that of the price-only slow memory from a kernel matched to the class horizon (eight hours). The class interval was narrower in 29 of 29 markets, by a median factor of 7.3 (smallest 2.6), and the price-only interval reached a bound of the fit in most. A registered secondary qualifies what this measures: corrected for the downward bias of day-demeaned regressions, the median class persistence was 1.0008 per five minutes, and below one in only 7 of 87 class fits. Positioning data restrict a persistence parameter that prices cannot bound; whether it reflects learning memory or slow turnover of positions, these data do not say.

7.4 The certificate and risk

In F1, a third registered primary asked whether the margin estimated on the previous 20 valid days forecasts the next valid day's largest adverse move out of sample, beyond realized variance, in a pooled model with symbol fixed effects and Clark–West loss differentials averaged within dates. The coefficient had the predicted sign (-1.13), but the test failed ($t = 1.51$, $p = 0.065$). For next-day realized variance, a registered secondary, the same design gave $t = 2.92$ ($p = 0.002$).

Two exploratory analyses on spent markets point the same way. In 7639 symbol-days, the elasticity of the next day's log adverse move with respect to $\log \hat{S}$, instrumented by its lag, was -0.13 (standard error 0.03), and of log variance -0.21 (0.04), against 1 and 2 in the positive class. On 10 October 2025, the day with the largest median adverse move, the Spearman correlation across 25 markets between the prior margin and the day's adverse move was +0.48. Post hoc, we first read this as Proposition 8 at work; Test R1 (Section 7.6) later found a positive elasticity for S on new markets with a shock control that excludes the estimated impact, which Pilot 7's control included, so the sign is more likely a property of that specification.

7.5 A registered test of fragility

Test S1 was registered on 25 October 2026, before the exploratory results above were seen, on 40 further markets and every day from January 2025 to September 2026 (34 evaluable). Two of its four primaries passed. A lower margin forecast higher next-day realized variance out of sample (coefficient -1.57, Clark–West $t = 4.82$, Holm-adjusted $p = 3 \times 10^{-6}$, gain 0.64 points of R^2 on 444 test dates), and the forecast was unchanged when the branching ratio of a Hawkes model of trade arrivals (Filimonov and Sornette, 2012; Hardiman et al., 2013), estimated by the INAR approximation of Kirchner (2016), was added to both models ($t = 4.84$); the branching ratio alone did not forecast variance ($t = -0.54$). Two failed. On the 20 worst market-wide days the prior margin did not rank which markets were hit hardest (mean partial rank correlation 0.000), and the forecast of next-day extreme moves fell short ($t = 1.54$, Holm $p = 0.12$). The margin carries information about the coming day's turbulence that the standard measure of self-excitation does not; it does not say where a crash will land.

7.6 What to certify instead

Test R1, registered on 25 October 2026 on 40 further markets (36 evaluable, none shared with S1), asked whether the estimated absolute response Ψ_H of Theorem 11, with H one hour, does what the margin did not. Within markets the next day's log adverse move rose with log Ψ_H, instrumented by its lag and controlling for shock size: elasticity 0.59 (standard error 0.09), positive as predicted, about half the theory's value of one; for variance the elasticity was 1.07 against two. Out of sample and beyond recent volatility, however, a larger Ψ_H forecast a smaller adverse move (coefficient -0.037), so the two registered forecasting primaries, which required a positive coefficient, failed. The data do not separate Ψ_H from the summed response: at a one-hour horizon the estimated responses rarely change sign (median $\Psi_H/|S_H| = 1.00$), the elasticity is the same in contrarian and momentum markets (0.59 and 0.60), and $S = \kappa/M$ itself has elasticity 0.46.

7.7 Real agents

H1 is prospective and uses participant-level data. Hyperliquid, a perpetual-futures exchange whose order book runs on a public blockchain, publishes every trade with the buyer's and the seller's wallet address; H1 was registered before any of its data existed, records every trade in 30 markets, and opens once at least 20 markets have 30 days of near-complete recording. For the ten largest accounts in each market it estimates each account's memory from its own position and asks whether two accounts have distinguishable memories in at least 75% of markets, and whether prices and anonymous flow fail to separate memories where the accounts' records succeed. It is the first test of the identification theory against real agents rather than proxies.

8 Discussion

The results separate two supervisory questions that are often run together. Whether a market is stable can be certified from public data, at a cost set by the market's distance from instability, without identifying any agent. Who is trading cannot be learned from market data at any comparable cost, and that is what disclosure is for. A supervisor whose goal is stability should therefore estimate the margin, ask for disclosures that bound memory or reveal total feedback, and treat identification as a separate, expensive task with its own justification.

Three limits are stated plainly. First, certification as proved here concerns stability. In the markets we study, where contrarian feedback is common, stability is far from worst-

case loss: the registered risk forecast failed, and the exploratory crash-day evidence went the wrong way. Worst-case loss over a horizon is certifiable from passive data through the absolute response (Theorem 11), but not from the margin, and whether that certificate forecasts losses is the subject of a registered test. Second, every certified market was far from the boundary; the evidence does not show that the procedure detects a market approaching instability. Third, the passive theorems need a memory bound, which Theorem 9 shows cannot be dropped, and the estimator in the tests pools a price and a flow equation that the theorems do not cover.

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A Proofs of Theorems 5 and 6

Theorem 5. (i) Both economies are $x_{t+1} = \rho_j x_t + e_{t+1}$ with $e_{t+1} \sim N(0, \kappa^2 s^2)$, $\rho_0 = 1-\mu$, $\rho_1 = 1+\mu$. The divergence is $\frac{1}{2}(2\mu)^2 \sum_{t<T} v_t$ with $v_0 = 0$, $v_{t+1} = \rho_0^2 v_t + 1$, so $v_t \le 1/(1-\rho_0^2) \le 1/\mu$ and the divergence is at most $2T\mu$. By Pinsker's inequality the total variation distance is at most $(T\mu)^{1/2}$, while a procedure with error at most α under both separates them with total variation at least $1-2\alpha$. (ii) For fixed $|\rho| < 1$, $\sqrt{T}(\hat\rho - \rho) \to N(0, 1-\rho^2)$ (Mann and Wald, 1943) and $1-\rho^2 = M(2-M)$. Along $\rho_T = 1 - M_T$ with $M_T \to 0$, $TM_T \to \infty$, Phillips and Magdalinos (2007) give $(T/2M_T)^{1/2}(\hat\rho - \rho_T) \to N(0,1)$; since $\hat M / M_T \to 1$ in probability, the studentized interval has asymptotic coverage $1-\alpha$ and excludes zero once $M > z_{1-\alpha/2}(2M/T)^{1/2}$.

Theorem 6. (i) g is real on $[0,1]$, equals one at zero and has no zero there, so $g(1) > 0$. Both economies have finite memory, since any real ϕ arises from $k_i = \phi_i/\kappa$, $i \ge 1$, and $a = \phi_0 - \kappa k_0$. The divergence is $\sum_{t<T} \mathbb{E}_+[((f_+ - f_-)(\mathcal{L})x_{t+1})^2]/(2\sigma^2)$, and $(f_+ - f_-)(z) = (\rho_- - \rho_+) z g(z)$, so it equals $(2\mu/g(1))^2 \sum_{t<T} \mathbb{E}_+ y_t^2/(2\sigma^2)$ with $y_t = g(\mathcal{L})x_t$. Under f_+, $y_{t+1} = \rho_+ y_t + e_{t+1}$ from $y_0 = 0$, so $\mathbb{E}_+ y_t^2 \le \sigma^2/(1-\rho_+^2) \le \sigma^2 g(1)/\mu$, and the divergence is at most $2T\mu/g(1)$. Pinsker's inequality finishes.

(ii) The regression is an invertible reparametrization of the stationary autoregression of x_{t+1} on $x_t, \dots, x_{t-p}$, so by Mann and Wald (1943) the coefficients are asymptotically normal

with covariance $\sigma^2\Gamma^{-1}$, and the entry for x_t is $1/\tau^2$, with τ^2 the residual variance of x_t after projection on the lagged differences. Each such residual is $h(\mathcal{L})x_t$ with h a polynomial of degree at most p and $h(1) = 1$. With $\delta = 1 - \rho$, $G = \max_{|z|=1} |g|$ and $|1 - \rho e^{-i\lambda}|^2 \le \delta^2 + \lambda^2$,

$$\tau^2 \ge \frac{\sigma^2}{2\pi G^2} \min_h \int_{-\pi}^{\pi} \frac{|h(e^{-i\lambda})|^2}{\delta^2 + \lambda^2}\, d\lambda.$$

Let $H(\lambda) = h(e^{-i\lambda})$ and I the integral. Since $\delta^2+\lambda^2 \le 1+\pi^2$, $\int |H|^2 \le (1+\pi^2)I$; by Cauchy–Schwarz $\max |H|^2 \le (p+1) \int |H|^2/(2\pi)$, and by Bernstein's inequality $\max |H'| \le p \max |H|$, so $\max |H'| \le pB\sqrt{I}$ with $B^2 = (p+1)(1+\pi^2)/(2\pi)$. With $r = \min(\pi, 1/(2pB\sqrt{I}))$, $|H| \ge 1/2$ on $|\lambda| \le r$, so $I \ge \arctan(r/\delta)/(2\delta)$. If $r \ge \delta$ this is at least $\pi/(8\delta)$; otherwise $r = 1/(2pB\sqrt{I}) < \delta$ and $I > 1/(4p^2B^2\delta^2)$. Hence $I \ge c/\delta$ with $c = 1/(4p^2B^2)$, $\tau^2 \ge \sigma^2 c/(2\pi G^2\delta)$, and with $\delta = M/g(1)$, $V = \sigma^2/\tau^2 \le 4p^2(p+1)(1+\pi^2)G^2M/g(1)$. Finally $g(z) = \prod_j (1 - z/r_j)$ with $|r_j| \ge 1 + \gamma$ gives $G \le 2^p$ and $g(1) \ge (\gamma/(1+\gamma))^p$.

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